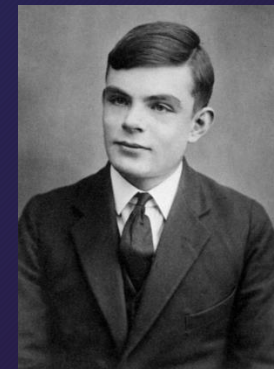
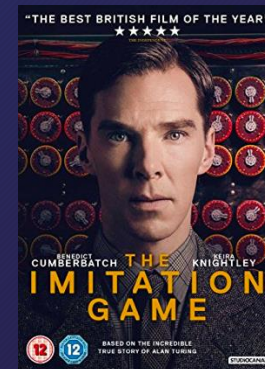


暖身-什麼是電腦？

2024 Fall

1-1 TURING MODEL

The idea of a universal computational device was first described by **Alan Turing** in 1937. He proposed that all computation could be performed by a special kind of a machine, now called a **Turing machine**. He based the model on the actions that people perform when involved in computation. He abstracted these actions into a model for a computational machine that has really changed the world.



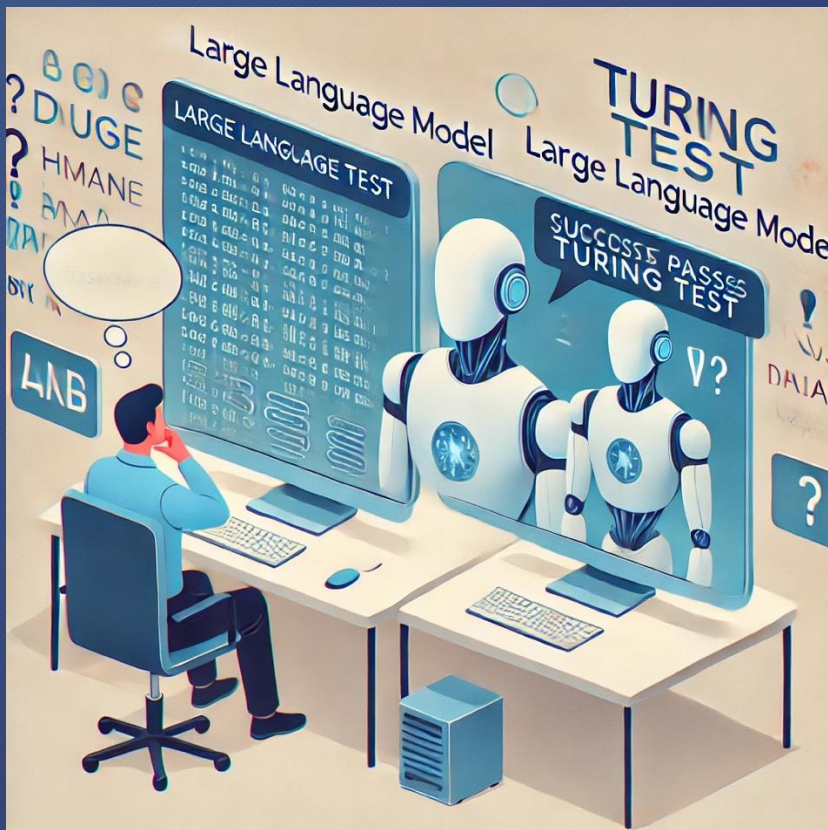
Turing Test



圖靈測試的主要目的是為了評估機器是否具有與人類相當的智能。

- 測試涉及三方:一名人類裁判、一名人類參與者和一台電腦。
- 人類裁判通過文字介面與人類參與者和電腦進行對話,但不知道哪個是人類,哪個是電腦。
- 在對話過程中,裁判可以問任何問題或討論任何話題。
- 如果經過一段時間的對話後,評判無法準確分辨出哪個是人類,哪個是電腦,那麼這台電腦就被認為通過了圖靈測試。

Thinking About



根據以上定義 Claude or ChatGPT4o 是否可能通過 Turing Test?

- 應該可以喔！
- Turing Test 只測試語言能力,可能忽視了智慧的其他方面能力。
- 模仿人類並不一定等同於真正的智慧。

Data processors

Before discussing the Turing model, let us define a computer as a data processor. Using this definition, a computer acts as a black box that accepts input data, processes the data, and creates output data (Figure 1.1). Although this model can define the functionality of a computer today, it is too general. In this model, a pocket calculator is also a computer (which it is, in a literal sense).



Figure 1.1 A single purpose computing machine

Programmable data processors

The **Turing model** is a better model for a general-purpose computer. This model adds an extra element to the specific computing machine: **the program**. A program is a set of instructions that tells the computer what to do with data. Figure 1.2 shows the Turing

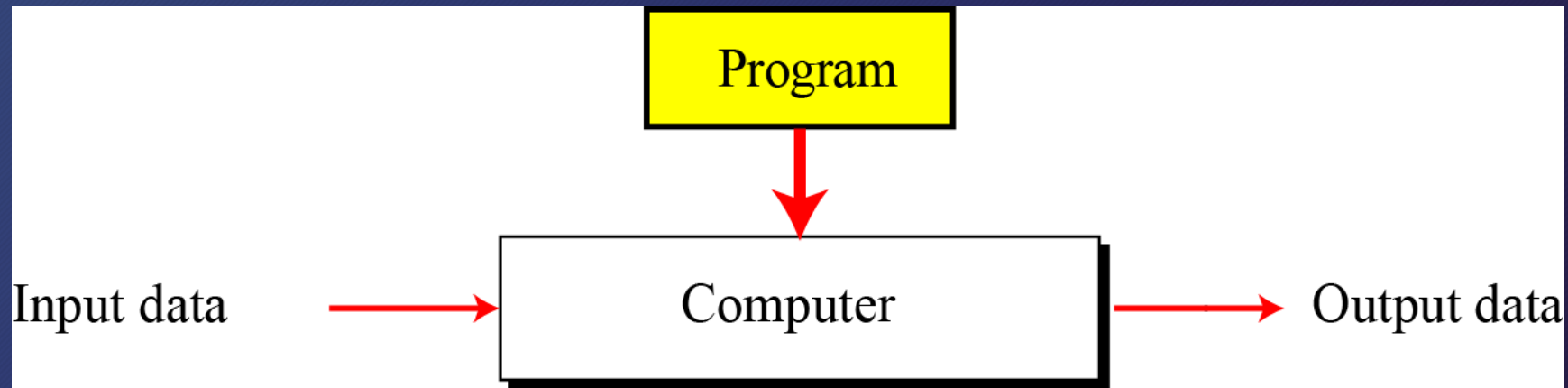


Figure 1.2 A computer based on the Turing model

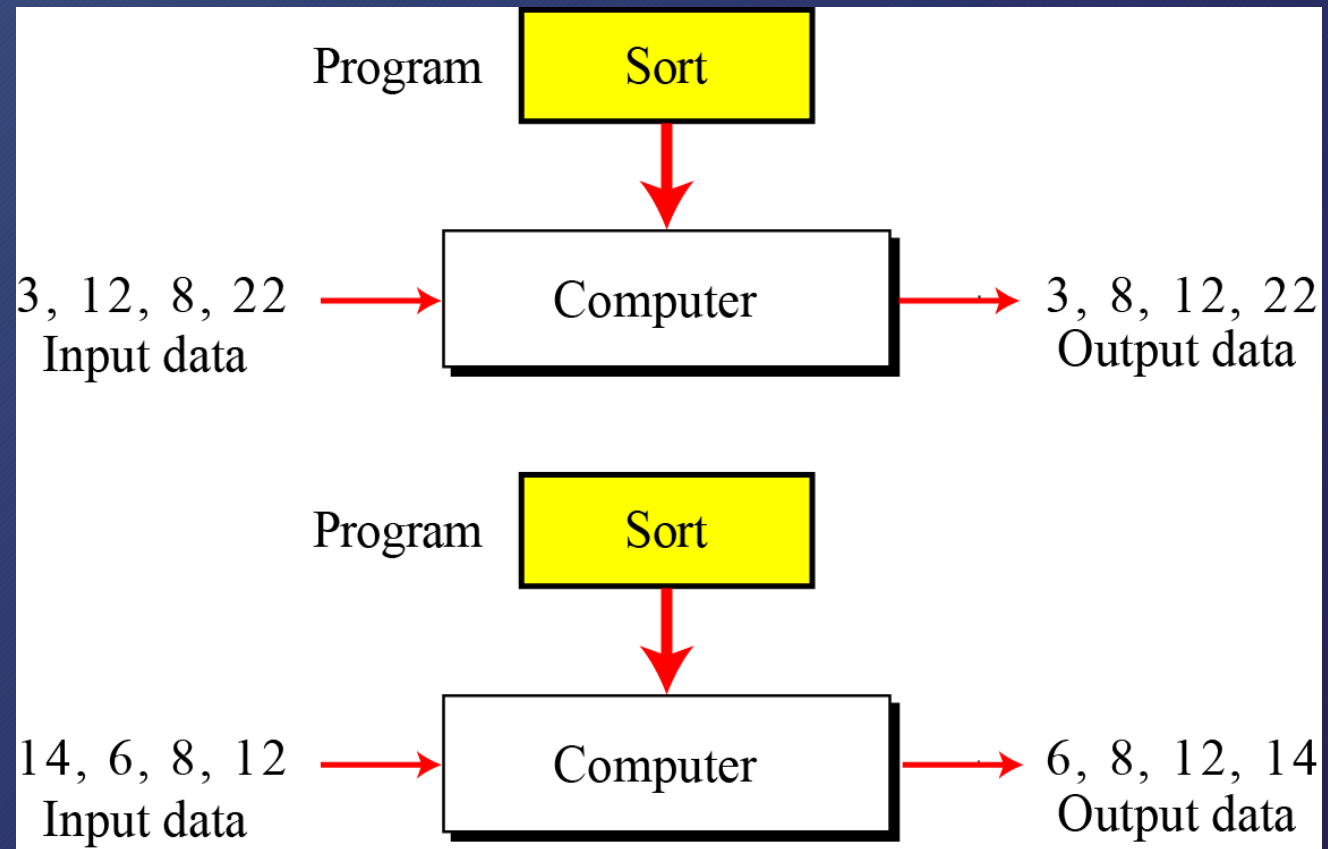


Figure 1.3 The same program, different data

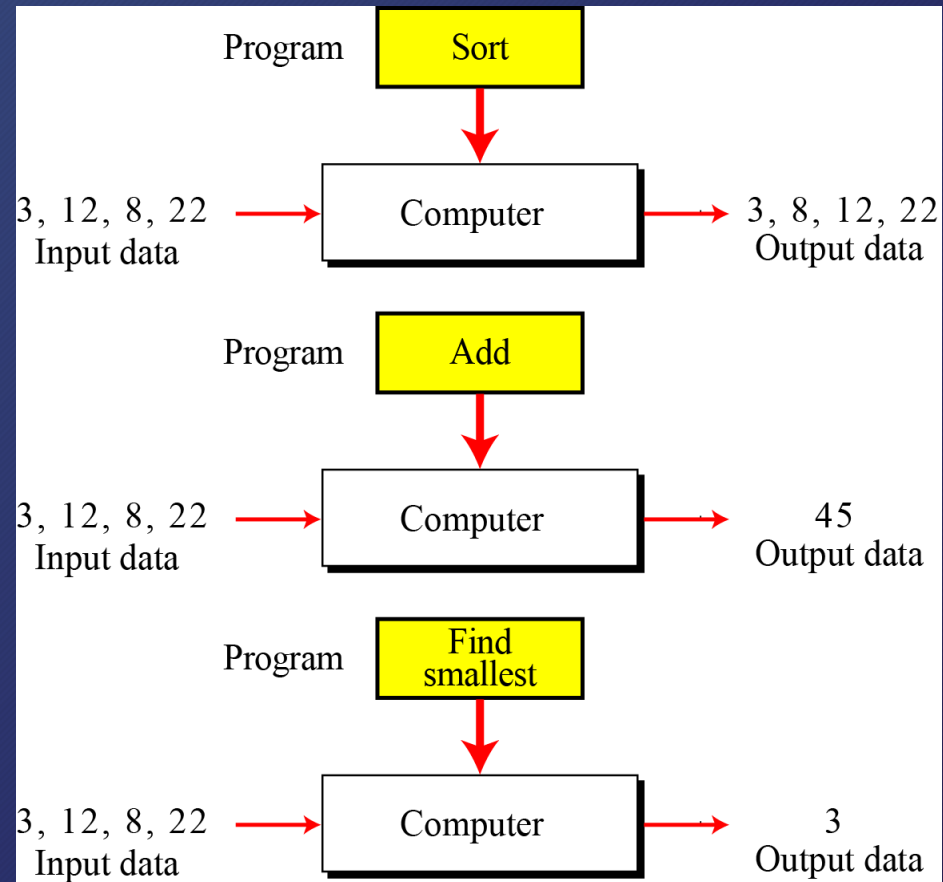


Figure 1.4 The same data, different programs

The universal Turing machine

Universal Turing Machine (UTM) 是一種理論模型，給予適當的 program 後，它可以執行任何 computation。這是最早描述現代 computer 的概念。可以證明，功能強大的 computer 與 UTM 具有相同的 computational 能力。兩者只需要 data 和 program——即 computation 的指令集——就能執行任何任務。本質上，UTM 可以計算任何 computable 的事情。

1-2 VON NEUMANN MODEL

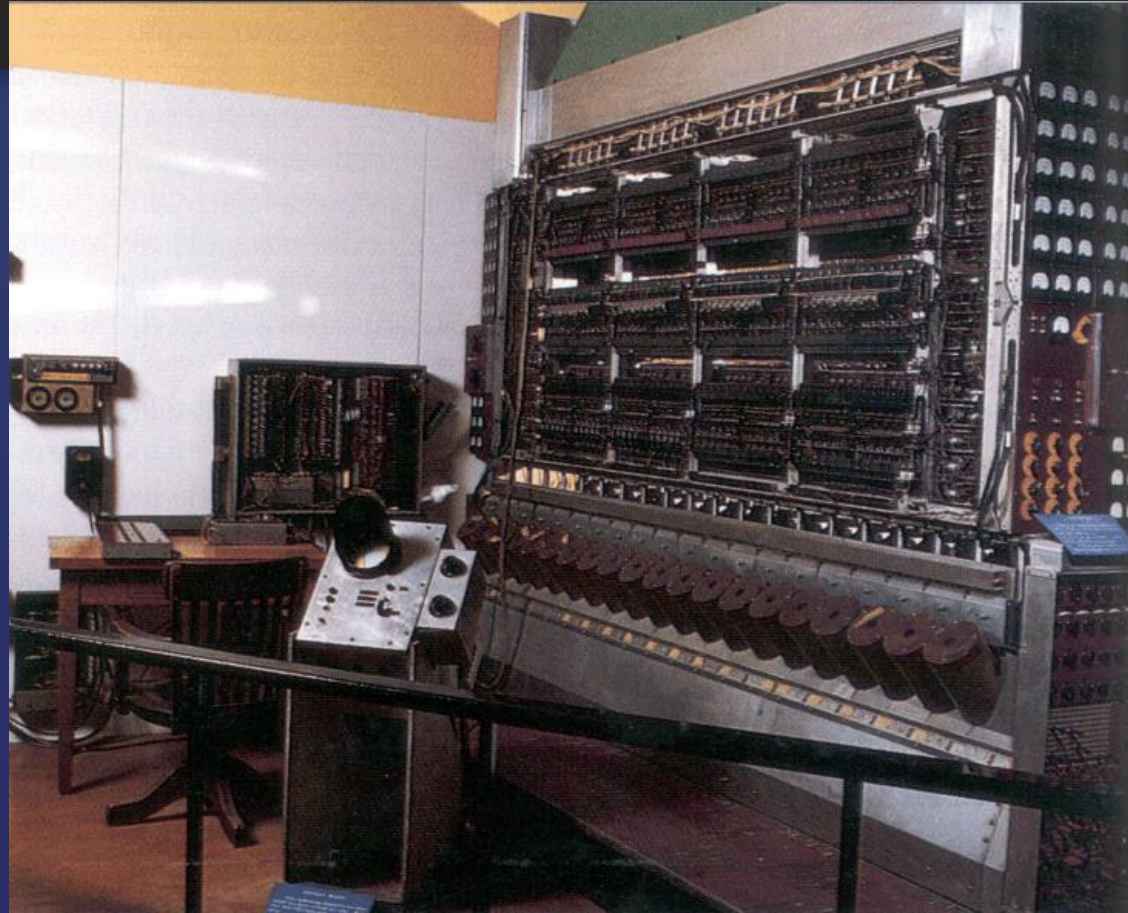


Von Neumann John

約翰·馮·諾伊曼（John von Neumann），現代計算機科學的奠基人之一，生於1903年，是匈牙利裔美國數學家。他提出了馮·諾伊曼架構，這種架構成為大部分現代計算機的基礎。除了計算機科學，他在數學、物理、經濟學等多個領域都有卓越貢獻。他參與了曼哈頓計劃，並對博弈論的發展產生了深遠影響。馮·諾伊曼的思想塑造了現代科技，為當代的計算機技術打下了堅實的基礎。

Three concept of Neumann's model

- Binary data
- Sequential execution
- The stored program concept



IAS computer (Institute of Advance Study , 就是高等研究院的名字)

Four subsystems

Computers built on the **von Neumann model** divide the computer hardware into four subsystems: **memory**, **arithmetic logic unit**, **control unit**, and **input/output** (Figure 1.5).

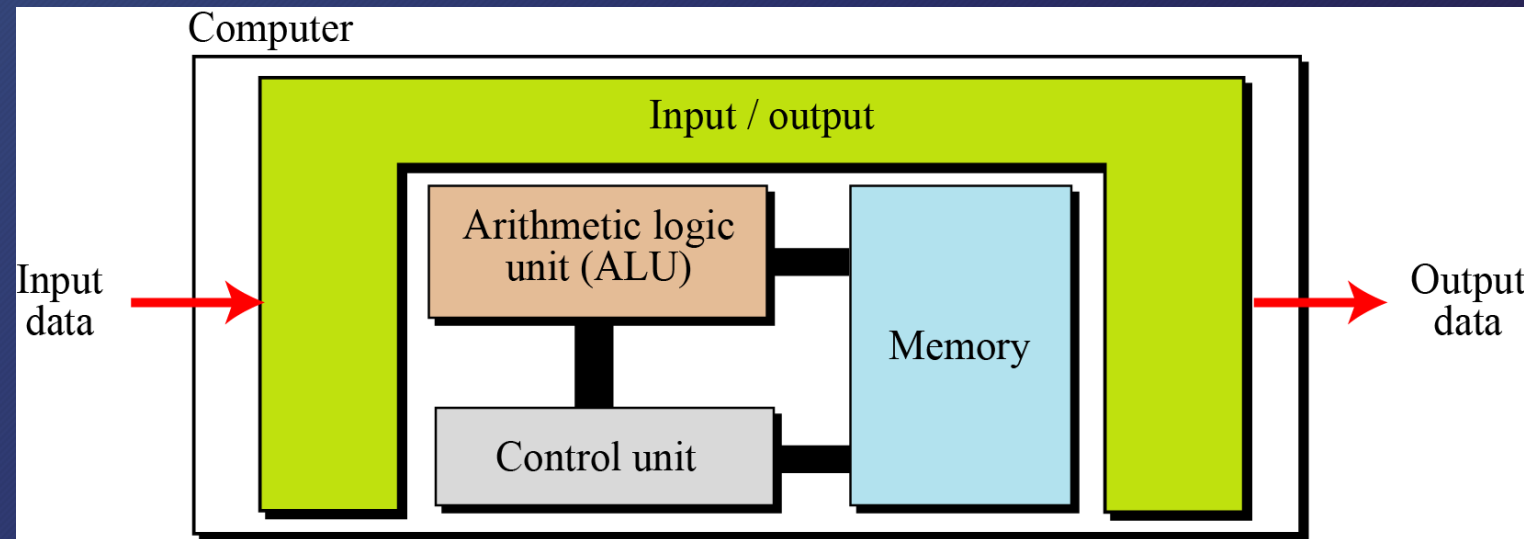


Figure 1.5 The von Neumann model

1-3 COMPUTER COMPONENTS

We can think of a computer as being made up of three components: (**computer hardware**) 有形, (**data**, and **computer software.**) 無形

Computer hardware

Computer hardware today has four components under the von Neumann model, although we can have different **types** of **memory**, different types of **input/output** subsystems, and so on. We discuss computer hardware in more detail in Chapter 5.

Data

The von Neumann model clearly defines a computer as a data processing machine that accepts the input data, processes it, and outputs the result.

Computer software

The main feature of the Turing or von Neumann models is the **concept of the program**. Although early computers did not store the program in the computer's memory, they did use the concept of programs. Programming those early computers meant changing the wiring systems or turning a set of switches on or off. Programming was therefore a task done by an operator or engineer **before the actual data processing began**.

<https://play.kahoot.it/v2/?quizId=c947b3da-347b-44fc-a4d6-1fc8abb01984>

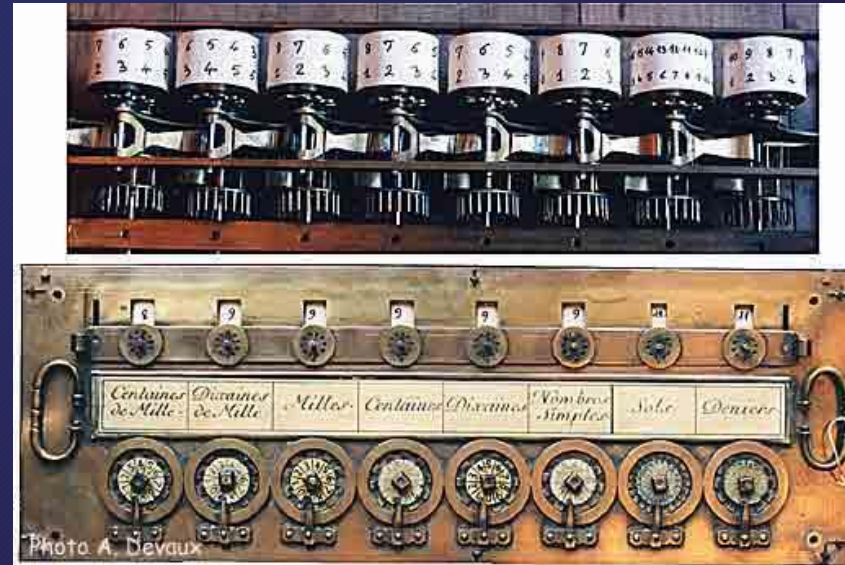
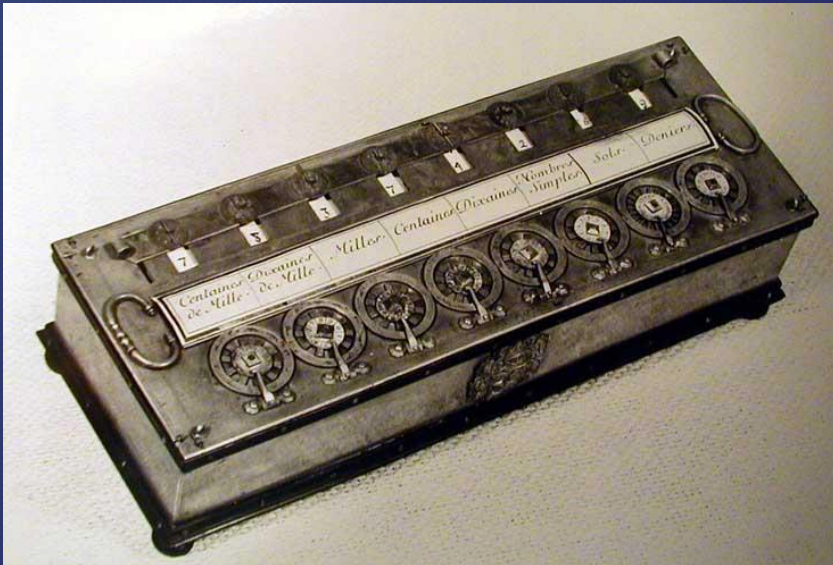
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Kahoot Time

Mechanical machines (before 1930)

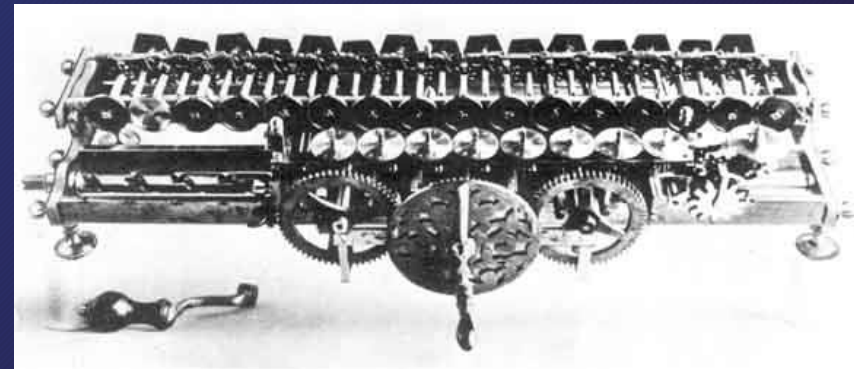
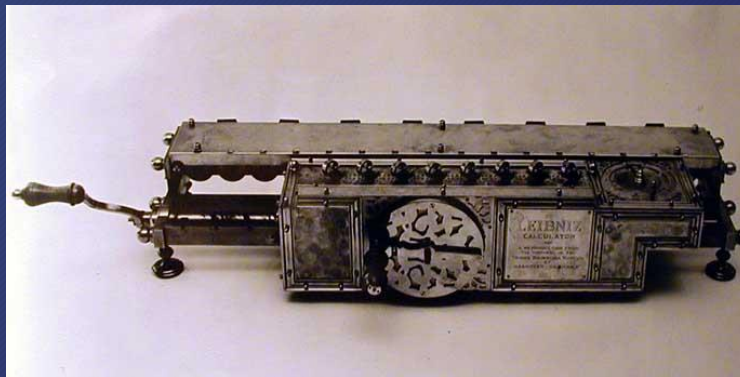
During this period, several computing machines were invented that bear little resemblance to the modern concept of a computer.

- ❑ In the 17th century, Blaise Pascal, a French mathematician and philosopher, invented **Pascaline**. (*for addition and subtraction*)



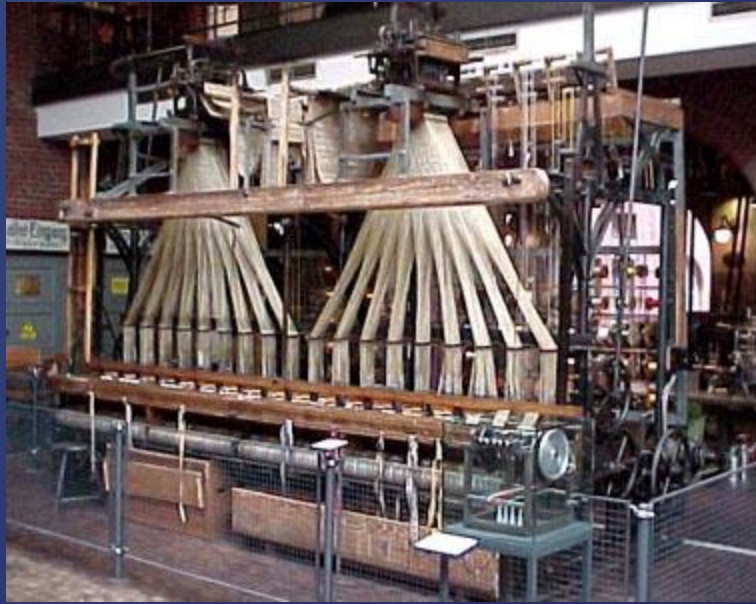
Mechanical machines (before 1930)

- ❑ In the late 17th century, a German mathematician called Gottfried Leibnitz invented what is known as Leibnitz' Wheel. (for multiplication and division)



Mechanical machines (before 1930)

- ❑ The first machine that used the idea of storage and programming was the Jacquard loom, invented by Joseph-Marie Jacquard at the beginning of the 19th century.



❑ In 1823, Charles Babbage invented the **Difference Engine**. Later, he invented a machine called the **Analytical Engine** that parallels the idea of modern computers.

❑ In 1890, **Herman Hollerith**, working at the US Census Bureau, designed and built a programmer machine that could automatically read, tally, and sort data stored on punched cards.

The birth of electronic computers (1930–1950)

Between 1930 and 1950, several computers were invented by scientists who could be considered the pioneers of the electronic computer industry.

Early electronic computers

The early computers of this period did not store the program in memory—all were programmed externally. Five computers were prominent during these years:

- ❑ ABC (線性方程組 Binary)
- ❑ Z1
- ❑ Mark I.
- ❑ Colossus
- ❑ ENIAC (Decimal 可程式化)

Computers based on the von Neumann model

The first computer based on von Neumann's ideas was made in 1950 at the University of Pennsylvania and was called **EDVAC**. At the same time, a similar computer called **EDSAC** was built by Maurice Wilkes at Cambridge University in England.

Computer generations (1950–present)

Computers built after 1950 more or less follow the von Neumann model. They have become faster, smaller, and cheaper, but the principle is almost the same. Historians divide this period into generations, with each generation witnessing some major change in hardware or software (but not in the model).

First generation

The first generation (roughly 1950–1959) is characterized by the emergence of commercial computers.

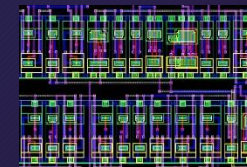
Second generation

Second-generation computers (roughly 1959–1965) used **transistors** instead of vacuum tubes. Two high-level programming languages, FORTRAN and COBOL, were invented and made programming easier.



Third generation

The invention of the **integrated circuit** reduced the cost and size of computers even further. Minicomputers appeared on the market. Canned programs, popularly known as **software packages**, became available. This generation lasted roughly from 1965 to 1975.



Fourth generation

The fourth generation (approximately 1975–1985) saw the appearance of microcomputers. The first desktop calculator, the Altair 8800, became available in 1975. This generation also saw the emergence of **computer networks**.

Fifth generation

This open-ended generation started in 1985. It has witnessed the appearance of **laptop** and **palmtop** computers, improvements in secondary storage media (**CD-ROM**, **DVD** and so on), the use of multimedia, and the phenomenon of virtual reality.

Extra Discussion

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- Microsoft and Bill Gates
- Apple and Steve Jobs
- Google and Larry Page, Sergey Brin
- Meta and Mark Zuckerberg
- Future Computer and Artificial Intelligence